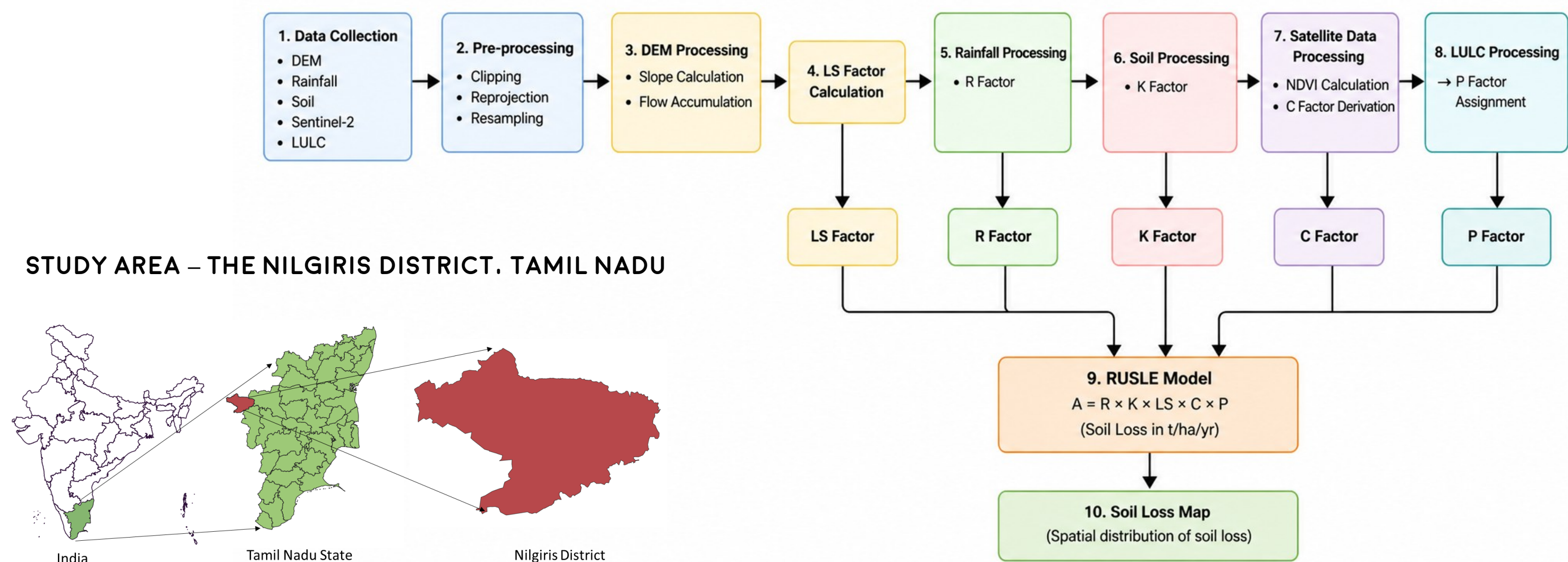


RUSLE-BASED SOIL EROSION ASSESSMENT FOR THE STUDY AREA

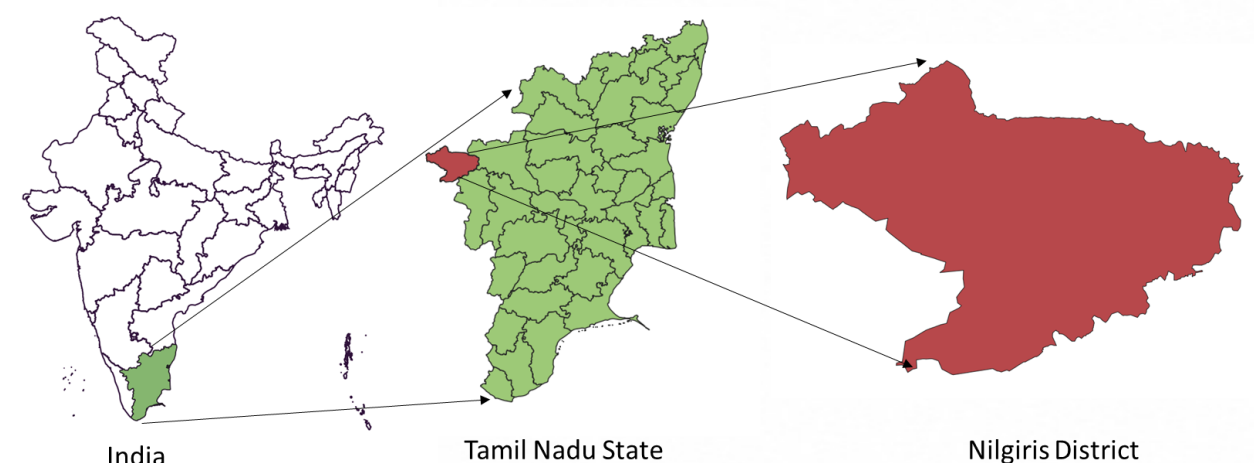
This study evaluates spatial patterns of soil erosion using the Revised Universal Soil Loss Equation (RUSLE). By integrating multi-source geospatial datasets including topography, rainfall, soil characteristics, vegetation cover, and land use practices, the analysis identifies erosion-prone zones and supports sustainable land management planning.

The workflow involves preprocessing of geospatial datasets followed by the derivation of five key RUSLE factors: rainfall erosivity (R), soil erodibility (K), topographic factor (LS), cover management (C), and conservation practice (P). These factors are spatially modelled and integrated to estimate annual soil loss and generate a comprehensive erosion risk map.

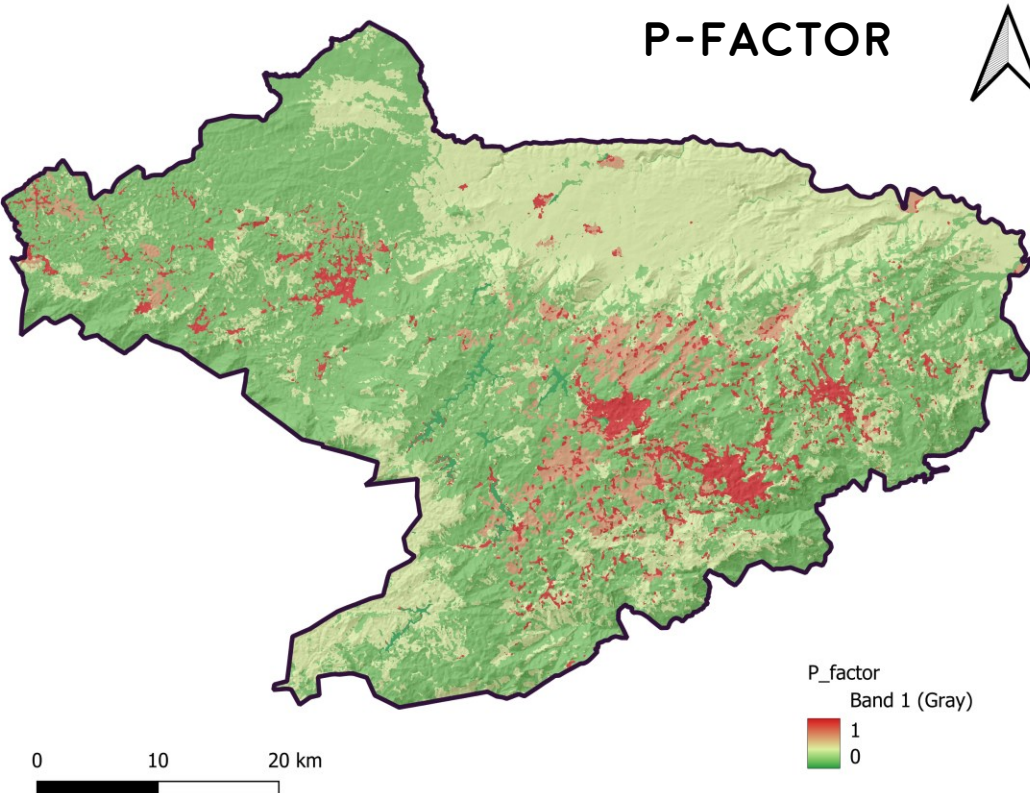
METHODOLOGICAL FRAMEWORK



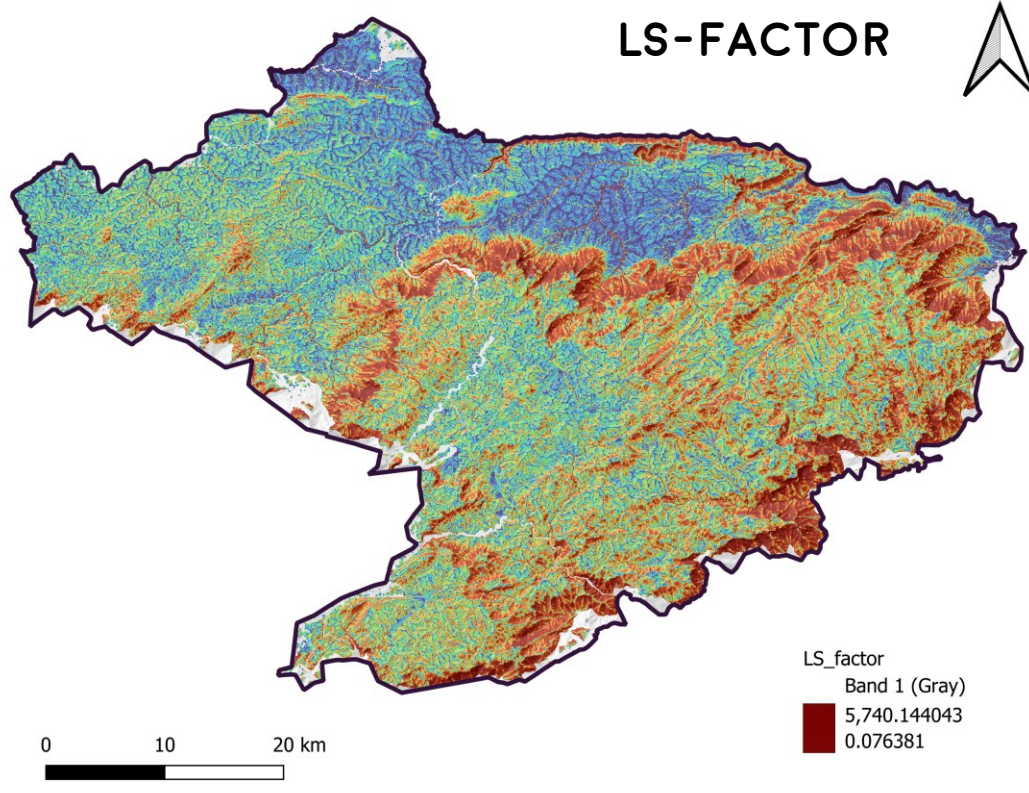
STUDY AREA – THE NILGIRIS DISTRICT, TAMIL NADU



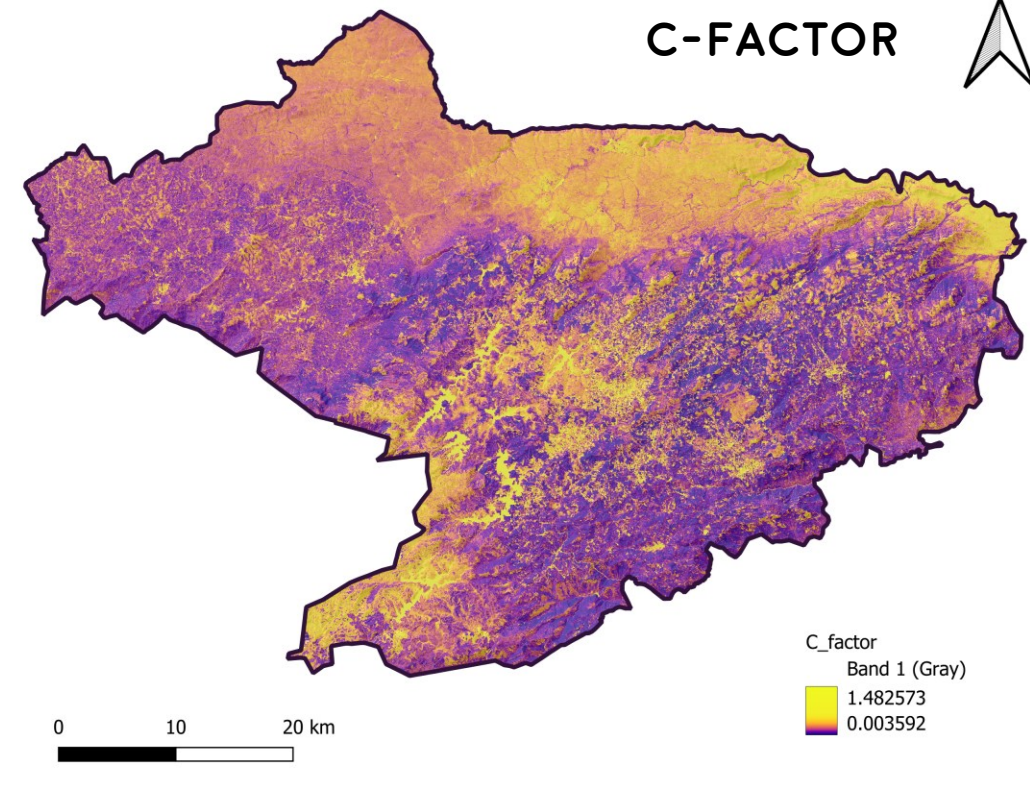
P-FACTOR



LS-FACTOR



C-FACTOR



RUSLE FRAMEWORK – REVISED UNIVERSAL SOIL LOSS EQUATION

